FineNet: A Joint Convolutional and Recurrent Neural Network Model to Forecast and Recommend Anomalous Financial Items

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ABSTRACT

Financial technology (FinTech) draws much attention in these years, with the advances of machine learning and deep learning. In this work, given historical time series of stock prices of companies, we aim at forecasting upcoming anomalous financial items, i.e., abrupt soaring or diving stocks, in financial time series, and recommending the corresponding stocks to support financial operations. We propose a novel joint convolutional and recurrent neural network model, *Financial Event Neural Network* (`FineNet`), to forecast and recommend anomalous stocks. Experiments conducted on the time series of stock prices of 300 well-known companies exhibit the promising performance of `FineNet` in terms of precision and recall. We build `FineNet` as a Web platform for live demonstration.

CCS CONCEPTS

• **Information systems → Data mining**.

KEYWORDS

CNN, RNN, Recommendation, Anomalous Financial items

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1 INTRODUCTION

Financial technology (FinTech) draws much attention in these years, with the advances of machine learning and deep learning. Forecasting time series of stock prices or returns is an essential task. Some study had used deep learning to predict the future values in financial time series [1]. Accurately predicting upcoming anomalous events of stocks can benefit stock recommendation for financial decision making, i.e., buying and selling operations, so that investors are able to get more profits and to avoid loss. While various financial analysis methods are used to support stock or capital recommendation [4], sentiment analysis [3] and investor-stock interactions [2] are also employed to enhance financial recommender, less attention is paid to investigate distill features from financial time series and to make recommendation.

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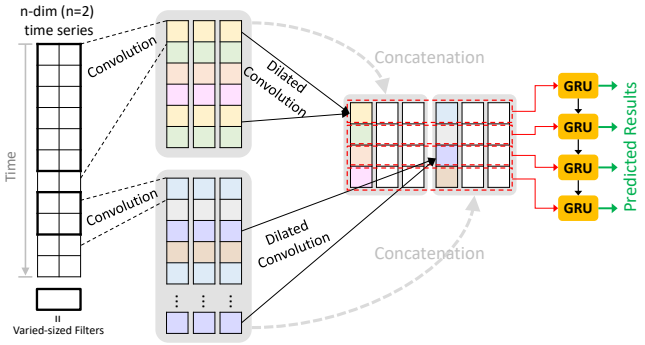

Figure 1: A simplified network architecture of `FineNet`.

This work is to forecast and recommend next anomalous financial events, i.e., abrupt soaring or diving of prices, in financial time series. That said, we aim at recommend most soaring and diving stocks to investors. Let R_Δ be the *return on investment* (ROI) of prices of time duration Δ, and let μ and σ be the mean and standard deviation for ROI values of a historical time series. Given the time series of stock prices of a company, we define an anomalous event as: if its ROI value is higher or lower than k standard deviation, i.e., $R_\Delta \geq \mu + k\sigma$ or $R_\Delta \leq \mu - k\sigma$, we say such time duration leads to an anomalous event Here we choose *weekly* ROI. Then given historical time series of stock prices of companies, our goal is to forecast anomalous events of stocks based on their predicted ROI values in the next week. Recommending upcoming anomalous stocks can help investors decide the buying and selling operations.

We propose a novel joint convolutional and recurrent neural network model, *Financial Event Neural Network* (`FineNet`), to forecast upcoming anomalous events. The idea is three-fold. First, we leverage convolutional neural network (CNN) to extract short-term significant features from time series. Second, we take advantage of the dilation convolution to learn long-term temporal features. Third, we utilize recurrent neural network (RNN) to further distill sequential features using the results from CNNs and make prediction. By ranking the stocks based on the forecasted values to their mean values in terms of the respective standard deviation, we can generate a ranking list of upcoming anomalous stocks for recommendation.

2 SKETCH OF THE PROPOSED `FINENET`

A simplified architecture of our `FineNet` is shown in Figure 1. The input is the n-dimensional *weekly* time series representing the

Table 1: Results in precision and recall by varying k.

	Precision			Recall		
	k=0.5	k=1.0	k=1.5	k=0.5	k=1.0	k=1.5
ARIMA	0.0098	0.0078	0.0060	0.0100	0.0067	0.0059
GRU	0.4729	0.0477	0.0035	0.0159	0.0112	0.0112
DRNN	0.5013	0.0097	0.0068	0.0350	0.0182	0.0112
FineNet	**0.6055**	**0.2982**	**0.0516**	**0.0665**	**0.0181**	**0.0124**

features of a company's stock prices, where each element is a feature value of the financial event of a week. We develop a set of different-sized filters to extract *short-term* features via the convolution operation. Then we use *dilated* convolution to learn *long-term* features of financial events, in which the dilation rate is also varied to capture different temporal effect. Equipped with long-term convolution results, we treat the derived values at the same time step as a feature vector. These vectors are fed into *Gated Recurrent Units* (GRUs) to learn sequential features from long-short term features, and the prediction results, i.e., the forecasted upcoming stock prices, can be obtained. Note that we use `ReLU` and `Linear` for the activation functions of CNNs and GRUs, respectively. For each stock, we calculate the difference of the forecasted value to the mean value in terms of respective standard deviation. Such values are used to generate the ranking lists of stocks. That is, we report the most soaring and diving stocks as the recommendation results.

`FineNet` allows two-fold flexibility. First, our model allows multivariate time series of features. In this work, we use only two basic features, the difference of stock values and the logarithmic difference of stock values. Common financial technical indices, such as moving average, stochastic oscillator and relative strength index, can be used as features. Second, to capture fine-grained temporal features, `FineNet` allows the multiple dilation convolution layers. It can be viewed as multi-resolution temporal feature extraction.

3 EXPERIMENTAL RESULTS

We collect time series of stock prices of 300 worldwide well-known companies from 2013-07 to 2018-11 using the API of Taiwan Economic Journal (TEJ) https://api.tej.com.tw/. By moving a fixed time point forwards, and setting different k (determining various anomalous degree), we have 159,685 normal and 197,915 anomalous events for $k = 0.5$, 262,559 normal and 95,041 anomalous events for $k = 1.0$, and 314,909 normal and 42,691 anomalous events for $k = 1.5$. We use Precision and Recall and as the evaluation metrics. The competing methods include Autoregressive Integrated Moving Average (ARIMA), Gated Recurrent Unit (GRU) and Denoising RNN model (DRNN) [1]. The experimental results are shown in Table 1. We can find the proposed `FineNet` significantly and consistently outperforms GRU and DRNN. The superiority of `FineNet` is more apparent when k is large. Such results imply that `FineNet` is able to not only predict regular ups and downs (smaller k), but also forecast extreme soaring or diving events (large k). In addition, it can be apparently found that precision values are significantly higher than recall values. In higher precision values, we can say those anomalous financial events forecasted by our `FineNet` are with high confidence. While it is difficult to predict and find all of anomalous events (quite low recall values), our `FineNet` is relatively good, comparing with the competing methods. In short, besides forecasting financial anomalous events, we believe `FineNet` can be also used for anomaly detection in other domains, such as traffic jam forecasting with traffic-flow time series, predicting air pollution with air-quality time series, and estimating disease risk with medical time series.

Figure 2: A system snapshot of `FineNet`.

4 SYSTEM DEMONSTRATION

A snapshot of our `FineNet` system[1] is displayed in Figure 2. The system consists of three main components. The first is allowing users to specify the stock, along with other settings, including the degree k of being an anomalous event, the number D of next days in the prediction, and top-N anomalous stocks in the recommendation list. When the user selects a certain stock, in the second part, we exhibit its historical time series of prices, and also show the predicted prices within the specified next D days. This is used to convince the users that the forecasting capability of our `FineNet`. Third, we report two lists of recommendation stocks based on the forecasted anomalous events. One is the stock list with most soaring prices in the forecasting, the other is the stock list with most diving prices. From such two lists, users could be able to make the most profitable decision in their financial operations.

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[1] Online system can be accessed via http://140.116.52.206/finenet/